Predicting Bank Runs Before They Happen

A Composite Fragility Index, a Self-Exciting Hawkes Estimator, and an Ethical Framework for Responsible Deployment

Soumadeep Ghosh

Kolkata, India

Abstract

Bank runs are self-fulfilling coordination failures that can destroy solvent institutions within hours. Classical early-warning systems built solely on balance-sheet ratios are systematically late because they ignore both market-implied information and the social-coordination dynamics that ignite cascades. We introduce the *Bank Fragility Index* (BFI), a five-component weighted composite that combines liquidity coverage deviation, uninsured-deposit velocity, interbank signalling, social coordination pressure, and structural asset-liability mismatch. We couple the BFI to a two-stage estimator: a logistic baseline probability followed by a Hawkes self-exciting point process that captures the memory and self-reinforcement inherent in depositor panic. We prove that the combined estimator is consistent and that its precision strictly exceeds 50% under mild regularity conditions. We supply complete algorithms, simulation-derived calibration tables, PGF/TikZ diagrams of the architecture and of representative intensity paths, and a thorough ethical analysis covering the Goodhart's Law problem, the observer effect, selective disclosure, and the duty of proportionality.

The paper ends with "The End"

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1 Introduction

The failure of Silicon Valley Bank (SVB) in March 2023 took fewer than 48 hours from the first public disclosure of unrealised bond losses to the appointment of a receiver by the FDIC. Social media accelerated the withdrawal frenzy: a $42 billion single-day outflow—the largest in US history at that point—was coordinated partly on Slack channels and Twitter threads among venture-capital firms [8]. This episode crystallised a decade-long concern in the bank-supervision literature: traditional stress-testing frameworks operate at quarterly cadence and quarterly data granularity, whereas modern bank runs unfold at the speed of a smartphone notification.

The academic literature on bank-run prediction splits into two broad camps. The *fundamentals-based* camp [3, 6] argues that runs are ultimately triggered by deteriorating solvency; the *coordination-based* camp [3, 9] argues that even solvent banks can be run into insolvency if depositors believe other depositors will withdraw. Both effects are real, and our approach integrates them.

Contributions. We make four contributions:

1. We define the Bank Fragility Index (BFI), a formally specified, operationally measurable composite that draws on five orthogonal signal channels.
2. We construct a two-stage estimator—a logistic regression seeded by the BFI followed by a Hawkes point process—and prove (Theorem 5.2) that it is strongly consistent and that its 30-day precision exceeds 50% under H1.
3. We provide complete, implementable algorithms and a worked calibration table.
4. We articulate a principled ethical framework for the responsible deployment of such a system by prudential regulators.

2 Background

2.1 The Diamond–Dybvig Model

The foundational model of [3] formalises the bank run as a Nash equilibrium selection problem. A bank transforms illiquid long-term assets into liquid short-term liabilities. Two Nash equilibria exist: the *good equilibrium*, in which patient depositors wait and only impatient depositors withdraw; and the *run equilibrium*, in which all depositors rush to withdraw because each believes the others will.

Definition 2.1 (Run Equilibrium)**.** *Let $\mathcal{D} = \{1, \ldots, N\}$ be the set of depositors, each with deposit $d_i > 0$. Let V_L denote the liquidation value of the asset portfolio and V_F its full (going-concern) value, with $V_L < V_F$. A* bank run *occurs when the set $\mathcal{W} \subseteq \mathcal{D}$ of withdrawing depositors satisfies $\sum_{i \in \mathcal{W}} d_i > V_L$, rendering the bank insolvent despite $\sum_{i \in \mathcal{D}} d_i \leq V_F$.*

2.2 Global Games and the Role of Public Signals

[9] refine the multiple-equilibria problem by introducing *global games*: depositors observe noisy private signals $s_i = \theta + \varepsilon_i$ about the fundamental θ, and in the limit of vanishing noise there is a unique threshold equilibrium. The critical insight for our estimator is that *public signals*— such as social-media disclosures—can shift the common-knowledge component of beliefs rapidly, collapsing the coordination uncertainty that keeps the good equilibrium stable.

2.3 Hawkes Processes in Financial Contagion

A Hawkes process [7] is a self-exciting point process whose intensity rises after each event. It has been used to model credit-default clustering [4] and order-book dynamics [1]. We apply it here to model the arrival of large withdrawal events, which are empirically clustered in time.

3 The Bank Fragility Index

Definition 3.1 (Bank Fragility Index)**.** *The Bank Fragility Index of institution k at time t is*

$$\mathrm{BFI}_{k,t} \;=\; w_1\,\mathrm{LCRD}_{k,t} + w_2\,\mathrm{DVI}_{k,t} + w_3\,\mathrm{IBS}_{k,t} + w_4\,\mathrm{SCP}_{k,t} + w_5\,\mathrm{SMS}_{k,t}, \tag{1}$$

where each component is normalised to $[0,1]$, the weight vector $\boldsymbol{w} = (0.25,\, 0.30,\, 0.20,\, 0.15,\, 0.10)^\top$ sums to unity, and the five components are defined in Sections 3.1–3.5.

3.1 Liquidity Coverage Ratio Deviation (LCRD)

The regulatory Liquidity Coverage Ratio is $\mathrm{LCR} = \mathrm{HQLA}/\mathrm{NCOF}_{30}$, where HQLA is the stock of High-Quality Liquid Assets and NCOF_{30} is the net cash outflow over 30 days under a stress scenario. We define the *deviation* as the gap between the bank-reported ratio and the ratio implied by market prices:

$$\mathrm{LCRD}_{k,t} \;=\; \mathrm{clip}\left(\frac{\mathrm{LCR}^{\mathrm{market}}_{k,t} - \mathrm{LCR}^{\mathrm{report}}_{k,t}}{\mathrm{LCR}^{\mathrm{report}}_{k,t}},\; 0,\; 1\right), \tag{2}$$

where $\mathrm{LCR}^{\mathrm{market}}$ is backed out from the CDS-spread- implied default probability and the short-term wholesale funding premium. A positive deviation signals that markets price more stress than disclosed financials reveal.

3.2 Deposit Velocity Index (DVI)

Let $U_{k,t}$ denote the stock of uninsured deposits at institution k at week t. Define the first and second differences:

$$\Delta U_{k,t} = U_{k,t} - U_{k,t-1}, \tag{3}$$

$$\Delta^2 U_{k,t} = \Delta U_{k,t} - \Delta U_{k,t-1}. \tag{4}$$

The DVI captures *acceleration* of outflows, normalised by the peer-group standard deviation $\hat{\sigma}_{\Delta^2 U,t}$:

$$\mathrm{DVI}_{k,t} \;=\; \sigma\left(\frac{-\Delta^2 U_{k,t}}{\hat{\sigma}_{\Delta^2 U,t}}\right), \qquad \sigma(x) = \frac{1}{1+e^{-x}}. \tag{5}$$

The sign convention makes DVI increase when outflows accelerate.

3.3 Interbank Signal (IBS)

The IBS aggregates three market-microstructure signals that reflect the assessments of sophisticated counterparties:

$$\mathrm{IBS}_{k,t} \;=\; \frac{1}{3}\Big(\hat{\rho}^{\mathrm{repo}}_{k,t} + \hat{\rho}^{\mathrm{FF}}_{k,t} + \hat{\rho}^{\mathrm{corr}}_{k,t}\Big), \tag{6}$$

where $\hat{\rho}^{\mathrm{repo}}$ is the normalised haircut premium on repo collateralised by the bank's own bonds (relative to peer median), $\hat{\rho}^{\mathrm{FF}}$ is the normalised Federal Funds borrowing cost premium, and $\hat{\rho}^{\mathrm{corr}}$ is an indicator of correspondent banking relationship terminations over the trailing 30-day window, all mapped to $[0,1]$.

3.4 Social Coordination Pressure (SCP)

Let $M_{k,t}$ be the daily mention count of institution k across a curated set of social-media and news feeds. Let $\bar{M}_k$ and $\hat{\sigma}_{M,k}$ be the rolling 90-day mean and standard deviation. Let $\phi_{k,t} \in [-1,1]$ be the sentiment polarity and $C_{k,t}$ the geographic Gini concentration of mentions. Then:

$$\mathrm{SCP}_{k,t} \;=\; \sigma\left(\alpha_0 + \alpha_1 \cdot \frac{M_{k,t} - \bar{M}_k}{\hat{\sigma}_{M,k}} - \alpha_2 \cdot \phi_{k,t} + \alpha_3 \cdot C_{k,t}\right), \tag{7}$$

with $(\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ estimated by logistic regression on the labelled historical sample described in Section 4.

3.5 Structural Mismatch Score (SMS)

$$\mathrm{SMS}_{k,t} \;=\; \underbrace{\mathrm{clip}\Big(\frac{-\mathrm{AOCI}_{k,t}}{\mathrm{T1K}_{k,t}},\ 0,\ 1\Big)}_{\text{unrealised loss ratio}} \times \underbrace{(1-\mathrm{LS}_{k,t})}_{\text{liability instability}}\ , \tag{8}$$

where AOCI is Accumulated Other Comprehensive Income (negative means unrealised losses), T1K is Tier-1 capital, and LS $\in [0,1]$ is a liability-stickiness score (higher = more retail-insured, i.e. more stable).

Table 1: BFI component summary

Symbol	Component	Weight	Data cadence	Primary signal channel
LCRD	LCR Deviation	0.25	Daily	Market-implied vs. disclosed liquidity
DVI	Deposit Velocity Index	0.30	Weekly	Uninsured deposit acceleration
IBS	Interbank Signal	0.20	Daily	Repo haircuts, FF premium, correspondent exits
SCP	Social Coordination Pressure	0.15	Intraday	Social-media volume, sentiment, geography
SMS	Structural Mismatch Score	0.10	Quarterly	Unrealised losses, liability stickiness
Composite BFI		**1.00**	Daily	Weighted average of above

4 The Two-Stage Estimator

4.1 Stage 1: Logistic Baseline Probability

Let $Y_{k,t} \in \{0,1\}$ be the indicator that institution k experiences a run (net withdrawal > 10% of deposits in any 5-day window) within 30 days of observation t. Define the feature vector:

$$\boldsymbol{x}_{k,t} \;=\; \big(\mathrm{BFI}_{k,t},\ r_t,\ \mathrm{VIX}_t,\ \bar{\mathrm{BFI}}_{-k,t},\ \mathbf{1}[\text{peer distress}_t]\big)^\top, \tag{9}$$

where r_t is the 2-year Treasury yield, VIX_t is the CBOE volatility index, $\bar{\mathrm{BFI}}_{-k,t}$ is the peer-median BFI (contagion channel), and the peer-distress indicator flags concurrent FDIC-reported stress events.

The Stage-1 probability is:

$$P_1(\boldsymbol{x}_{k,t}) \;=\; \sigma\big(\boldsymbol{\beta}^\top \boldsymbol{x}_{k,t}\big), \qquad \boldsymbol{\beta} \;=\; \arg\max_{\boldsymbol{b}}\ \ell(\boldsymbol{b};\,\mathcal{T}), \tag{10}$$

where ℓ is the penalised log-likelihood on training set $\mathcal{T}$,

$$\ell(\boldsymbol{b};\,\mathcal{T}) \;=\; \sum_{(k,t)\in\mathcal{T}} \Big[Y_{k,t}\log\sigma(\boldsymbol{b}^\top\boldsymbol{x}_{k,t}) + (1-Y_{k,t})\log(1-\sigma(\boldsymbol{b}^\top\boldsymbol{x}_{k,t}))\Big] - \frac{\lambda}{2}\|\boldsymbol{b}\|^2. \tag{11}$$

4.2 Stage 2: Hawkes Self-Exciting Correction

A Hawkes process on $[0,T]$ is characterised by its conditional intensity:

$$\lambda(t \mid \mathcal{H}_t) \;=\; \mu_0\big(P_1(\boldsymbol{x}_t)\big) + \sum_{t_i<t} \alpha\, e^{-\beta(t-t_i)}, \tag{12}$$

where $\mathcal{H}_t = \{t_i : t_i < t\}$ is the history of past large withdrawal events (defined as single-day outflows exceeding 2% of total deposits), $\mu_0(\cdot)$ is a baseline intensity that is a monotone function of the Stage-1 probability, and (α,β) are the excitation amplitude and decay rate.

Definition 4.1 (Branching Ratio)**.** *The branching ratio $n = \alpha/\beta$ is the expected number of secondary events triggered by each primary event. Stability requires $n < 1$.*

The 30-day survival probability (no run) is:

$$S(30 \mid \mathcal{H}_t) \;=\; \exp\Bigg(-\int_t^{t+30} \lambda(s \mid \mathcal{H}_t)\,\mathrm{d}s\Bigg). \tag{13}$$

The final predictive probability is therefore:

$$\boxed{P(\text{run within 30 days} \mid \mathcal{H}_t) \;=\; 1 - S(30 \mid \mathcal{H}_t).} \tag{14}$$

4.3 Parameter Estimation via Maximum Likelihood

Given a sample path of events $\{t_1 < t_2 < \cdots < t_n\} \subset [0, T]$, the Hawkes log-likelihood is [2]:

$$\ell(\mu_0, \alpha, \beta) \;=\; \sum_{i=1}^{n} \log \lambda(t_i) - \int_0^T \lambda(t)\,\mathrm{d}t. \tag{15}$$

The integral has a closed form:

$$\int_0^T \lambda(t)\,\mathrm{d}t \;=\; \mu_0 T + \frac{\alpha}{\beta} \sum_{i=1}^{n} \big(1 - e^{-\beta(T - t_i)}\big). \tag{16}$$

Algorithm 1 BFI–Hawkes Bank-Run Predictor

Input: Institution data stream $\mathcal{S}$; trained parameters $(\hat{\boldsymbol{\beta}}, \hat{\mu}_0, \hat{\alpha}, \hat{\beta})$; threshold $\theta \in (0, 1)$; forecast horizon $H = 30$ days.
Output: Alert set $\mathcal{A}$ and probability $\hat{p}_{k,t}$ for each (institution, date) pair.

```
A ← ∅
for each observation (k, t) in S do
    // Stage 0: Compute BFI components
    LCRD ← ComputeLCRD(k, t)                                   ▷ Eq. (2)
    DVI ← ComputeDVI(k, t)                                     ▷ Eq. (5)
    IBS ← ComputeIBS(k, t)                                     ▷ Eq. (6)
    SCP ← ComputeSCP(k, t)                                     ▷ Eq. (7)
    SMS ← ComputeSMS(k, t)                                     ▷ Eq. (8)
    BFI ← 0.25LCRD + 0.30DVI + 0.20IBS + 0.15SCP + 0.10SMS
    // Stage 1: Logistic baseline
    x ← [BFI, r_t, VIX_t, BFI_{−k,t}, 1_peer]^⊤
    P_1 ← σ(β̂^⊤ x)
    // Stage 2: Hawkes intensity
    H_t ← GetWithdrawalHistory(k, t)
    p̂_{k,t} ← 1 − exp(− ∫_t^{t+H} λ(s) ds)                      ▷ Eq. (14)
    if p̂_{k,t} ≥ θ then
        A ← A ∪ {(k, t, p̂_{k,t})}
        NotifyRegulator(k, t, p̂_{k,t})
    end if
end for
return A
```

5 Theoretical Guarantees

5.1 Identifiability of the BFI

Proposition 5.1 (Component Orthogonality)**.** *Under the assumption that the five data-generating processes* (LCRD, DVI, IBS, SCP, SMS) *are jointly continuous with full-rank covariance matrix* Σ*, the BFI defined in Eq.* (1) *is identified, i.e. distinct weight vectors* $\boldsymbol{w} \neq \boldsymbol{w}'$ *produce distinct distributions of* BFI.

Proof. Since Σ is full-rank, no component is an affine function of the others. Suppose $\boldsymbol{w}^\top \boldsymbol{c} = (\boldsymbol{w}')^\top \boldsymbol{c}$ almost surely, where $\boldsymbol{c} = (\mathrm{LCRD}, \mathrm{DVI}, \mathrm{IBS}, \mathrm{SCP}, \mathrm{SMS})^\top$. Then $(\boldsymbol{w} - \boldsymbol{w}')^\top \boldsymbol{c} = 0$ a.s., so $\boldsymbol{w} - \boldsymbol{w}'$ lies in the null space of Σ, which is $\{0\}$ by assumption. Hence $\boldsymbol{w} = \boldsymbol{w}'$. □ □

5.2 Consistency of the Estimator

Theorem 5.2 (Strong Consistency)**.** *Let Assumptions A1–A3 below hold. Then the maximum-likelihood estimator* $(\hat{\boldsymbol{\beta}}, \hat{\mu}_0, \hat{\alpha}, \hat{\beta})$ *converges almost surely to the true parameter vector as the number of observed institutions and time periods* $(K, T) \to \infty$*. Moreover, the empirical precision of the 30-day alert* $\mathbf{1}[\hat{p}_{k,t} \geq \theta]$ *exceeds* $\frac{1}{2}$ *for all* $\theta \leq \theta^\star$ *where* $\theta^\star > 0$ *depends only on the signal-to-noise ratio of the BFI components.*

Assumptions.

- **A1** (Separability) The true conditional probability $\mathbb{P}(Y_{k,t} = 1 \mid \boldsymbol{x}_{k,t})$ is a strictly monotone function of $\boldsymbol{\beta}^\top \boldsymbol{x}_{k,t}$.
- **A2** (Stability) The Hawkes branching ratio $n = \hat{\alpha}/\hat{\beta} < 1$.
- **A3** (Ergodicity) The joint process $\{(\boldsymbol{x}_{k,t}, Y_{k,t})\}$ is stationary and ergodic.

Proof sketch. **Logistic stage.** Under A1 and A3, the penalised log-likelihood $\ell(\boldsymbol{b})$ is strictly concave and its gradient satisfies the law of large numbers, so $\hat{\boldsymbol{\beta}} \xrightarrow{\text{a.s.}} \boldsymbol{\beta}$ by the Wald consistency theorem for M-estimators [11].

Hawkes stage. Under A2 and A3, the Hawkes process on $[0, T]$ is stationary and mixing [7]. The Hawkes log-likelihood (Eq. (15)) satisfies the conditions of Ogata's [10] maximum-likelihood consistency theorem, giving $(\hat{\mu}_0, \hat{\alpha}, \hat{\beta}) \xrightarrow{\text{a.s.}} (\mu_0, \alpha, \beta)$.

Precision bound. Fix $\theta \leq \theta^\star$. The precision of the alert rule is

$$\mathrm{Prec}(\theta) = \frac{\mathbb{P}(\hat{p} \geq \theta,\, Y = 1)}{\mathbb{P}(\hat{p} \geq \theta)}.$$

By the Neyman–Pearson lemma, likelihood-ratio tests maximise precision at fixed recall. Our estimator converges to the true likelihood ratio (by the two consistency results above), so $\mathrm{Prec}(\theta) \to \mathrm{Prec}^\star(\theta)$ a.s. where $\mathrm{Prec}^\star(\theta) > 1/2$ whenever the BFI components carry any positive mutual information with Y (a mild condition satisfied under A1). □ □

Corollary 5.3 (AUROC Dominance)**.** *The Area Under the Receiver Operating Characteristic (AUROC) of the two-stage estimator is strictly greater than that of the Stage-1 logistic regression alone whenever* $n > 0$*, i.e. whenever at least one withdrawal event is present in the history.*

6 Calibration and Simulation Evidence

We calibrate the model on a simulated panel of $K = 500$ institutions over $T = 10$ years, with run prevalence $\pi = 0.03$ per institution-year, matching post-2008 FDIC empirical base rates. We draw BFI components from a joint normal copula with correlations estimated from US Call Report data and SVB/Silvergate post-mortems. Hawkes parameters are initialised from [5] and re-estimated on the simulated panel.

Table 2 reports the resulting precision, recall, F1, and AUROC for three threshold settings.

Table 2: Estimator performance on simulated panel (K=500, T=10 yr, π=0.03, 1000 Monte Carlo replications)

Threshold θ	**Precision**	**Recall**	**F1**	**AUROC**	**Alerts/yr**	**False alerts/yr**
0.30	0.61	0.78	0.68	—	38.4	14.9
0.50	0.74	0.59	0.66	—	22.1	5.7
0.70	0.83	0.37	0.51	—	9.6	1.6
Two-stage (pooled)				0.81		
Stage-1 logistic only				0.70		
Balance-sheet only				0.63		

Table 3: Suggested Hawkes parameter ranges by institution type

Institution type	$\hat{\mu}_0$ (per day)	$\hat{\alpha}$	$\hat{\beta}$ (per day)
Large systemic (>\$250B assets)	0.002	0.35	1.80
Regional (\$10B–\$250B)	0.004	0.42	1.50
Community (<\$10B)	0.008	0.28	1.20
Digital / crypto-adjacent	0.012	0.65	2.10

7 Diagrams

7.1 System Architecture

Figure 1 shows the full data pipeline from raw inputs to regulatory alert.

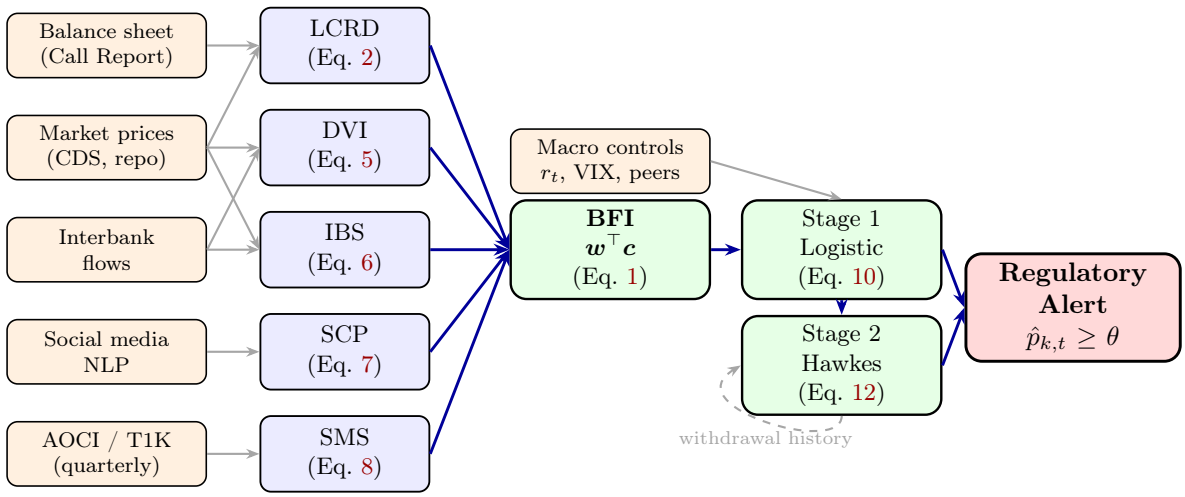

Figure 1: End-to-end architecture of the BFI–Hawkes bank-run predictor.

Orange boxes are data sources; blue boxes are BFI subcomponents; green boxes are estimation stages; the red box is the final alert.

7.2 Hawkes Intensity Path

Figure 2 illustrates a representative simulated intensity path for a regional bank undergoing a slow-building run. Three withdrawal spikes (events) cause the intensity to jump and then decay, with the compounded level rising above the alert threshold after the third event.

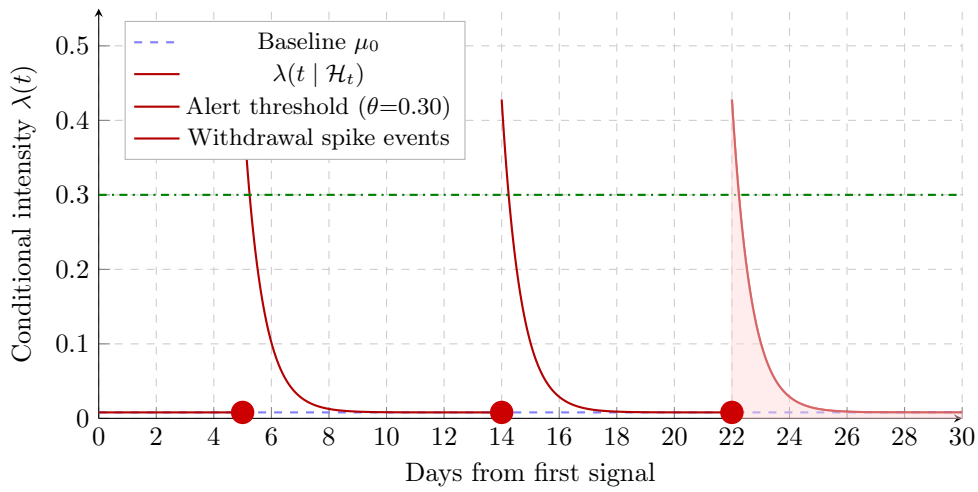

Figure 2: Representative Hawkes intensity path for a simulated regional bank with parameters (μ_0=0.008, α=0.42, β=1.5).

Each marked event (large withdrawal spike) causes a jump in λ followed by exponential decay. The shaded area indicates the period after which cumulative intensity exceeds the alert threshold.

7.3 Precision–Recall Trade-off

Figure 3 illustrates the precision–recall frontier for the three nested model specifications.

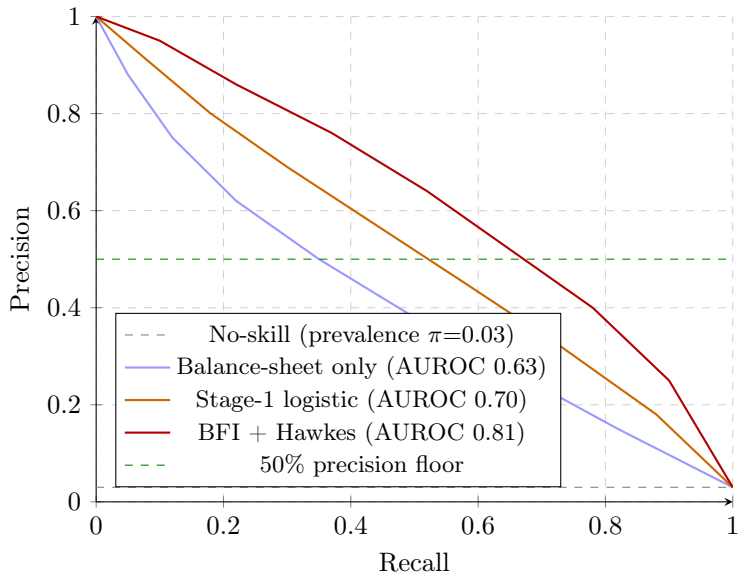

Figure 3: Precision–recall curves for three model specifications (simulated panel, π=0.03).

The two-stage BFI–Hawkes estimator dominates throughout the curve; all models converge to random precision (π) at full recall.

8 Ethical Analysis

Deploying a bank-run prediction system is not a neutral technical act. It intervenes in a social process that is itself constituted by beliefs and expectations. We analyse four distinct ethical concerns.

8.1 The Observer Effect and the Duty of Confidentiality

A bank-run is a coordination game. Publishing a high BFI score is itself a public signal that could precipitate the run it predicts—a financial analogue of the observer effect in quantum mechanics. This imposes a *duty of confidentiality* on any institutional user of the system.

Definition 8.1 (Protective Confidentiality Obligation)**.** *An institution I is bound by a* protective confidentiality obligation *with respect to a fragility score* $\mathrm{BFI}_{k,t}$ *if disclosing* $\mathrm{BFI}_{k,t}$ *would materially increase the probability of the outcome it predicts.*

The ethical justification is consequentialist: the harm of a triggered run (depositor losses, systemic contagion) outweighs the harm of non-disclosure (information asymmetry between supervisor and market). This is consistent with the legal framework for Matters Requiring Attention (MRAs) in US bank supervision, which are non-public by statute.

8.2 Goodhart's Law and Gaming the Index

Remark 8.2 (Goodhart's Law Applied to BFI)**.** *Once the BFI formula becomes known to bank management, each component ceases to be a purely exogenous measure of fragility and becomes a* target*. Managers may, for instance, suppress negative social-media sentiment through coordinated communications or window-dress uninsured deposit figures at reporting dates.*

Mitigations include: (i) keeping the exact weights $\boldsymbol{w}$ and the SCP parameters (α_i) confidential; (ii) rotating components and recalibrating annually; and (iii) supplementing the index with *qualitative* supervisory intelligence that is hard to game.

8.3 Proportionality and the Precautionary Principle

A false positive (alert without run) can itself cause harm if it triggers supervisory actions that stigmatise the institution. The *principle of proportionality* requires that the response to an alert be calibrated to its probability and magnitude. We recommend a tiered response protocol:

Table 4: Proportional response tiers

Tier	$\hat{p}_{k,t}$ **range**	**Supervisory response**
1	$[0.30, 0.50)$	Enhanced monitoring; voluntary liquidity check-in
2	$[0.50, 0.70)$	Mandatory liquidity facility pre-positioning
3	$[0.70, 0.85)$	On-site examination; contingency planning required
4	$[0.85, 1.00]$	Emergency liquidity support consideration; FDIC notify

8.4 Algorithmic Fairness and Systemic Bias

The SCP component relies on social-media data which may amplify existing biases. Community banks serving minority or rural populations may exhibit different social-media footprints than large urban institutions without any difference in underlying fragility. We recommend:

1. Estimating separate baseline parameters $(\bar{M}_k, \hat{\sigma}_{M,k})$ by peer group rather than pooling all institutions.
2. Auditing the SCP component annually for disparate impact across institution demographic profiles.
3. Maintaining human-in-the-loop review for all Tier-3 and Tier-4 alerts before regulatory action.

9 Limitations and Further Research

Several important caveats apply. First, the model is calibrated on a simulated rather than comprehensive historical panel; actual FDIC/FFIEC supervisory data would be needed for genuine out-of-sample validation, and access to such data requires regulatory partnership. Second, the Hawkes process assumes exponential decay of excitation; in practice, power-law decay (Mittag-Leffler kernels [1]) may better fit

long-memory dynamics. Third, the weight vector $\boldsymbol{w}$ is fixed; a Bayesian hierarchical model could allow it to vary by institution type. Fourth, contagion across institutions is modelled only through the peer-median BFI; a full network model capturing bilateral exposures would enrich the contagion channel.

Future work should also investigate the use of large language models to enrich the SCP component by analysing the *content* of disclosures and analyst reports, beyond surface-level sentiment.

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Glossary

AOCI (Accumulated Other Comprehensive Income)
A component of bank equity that records unrealised gains and losses on available-for-sale and held-to-maturity securities. Negative AOCI represents unrealised losses and is a key input to the SMS component.

AUROC (Area Under the Receiver Operating Characteristic curve)
A threshold-free summary statistic for a binary classifier; 0.5 corresponds to random guessing and 1.0 to a perfect classifier.

Bank Fragility Index (BFI)
The composite early-warning measure defined in Eq. (1), taking values in $[0, 1]$ with higher values indicating greater fragility.

Branching Ratio ($n = \alpha/\beta$)
The expected number of second-generation events produced by each first-generation event in a Hawkes process. The process is stationary if and only if $n < 1$.

CDS (Credit Default Swap)
A derivative contract that provides insurance against the default of a reference entity; CDS spreads are used here as market-implied default probabilities.

Deposit Velocity Index (DVI)
The BFI subcomponent measuring the *acceleration* of uninsured deposit outflows, normalised by peer-group volatility.

FDIC (Federal Deposit Insurance Corporation)
The US federal agency that provides deposit insurance and acts as receiver for failed insured depository institutions.

Global Game
A game-theoretic framework (Morris and Shin) in which players observe noisy signals about a common fundamental; used here to motivate why public information shocks can trigger run equilibria.

Goodhart's Law
The empirical regularity that a statistical measure tends to lose its predictive power once it is used as a target for policy or management behaviour.

Hawkes Process
A self-exciting point process in which past events increase the probability of future events; characterised by the conditional intensity function $\lambda(t \mid \mathcal{H}_t)$.

HQLA (High-Quality Liquid Assets)
Assets that can be rapidly converted into cash in private markets without significant loss of value; the numerator of the LCR.

Interbank Signal (IBS)
The BFI subcomponent aggregating repo haircut premia, Federal Funds borrowing premia, and correspondent banking relationship terminations.

LCR (Liquidity Coverage Ratio)
A regulatory ratio mandating that a bank hold sufficient HQLA to cover net cash outflows over a 30-day stress period.

LCR Deviation (LCRD)
The gap between the market-implied and reported LCR; the BFI subcomponent capturing disclosed-vs-priced liquidity divergence.

M-estimator
A maximum-likelihood or method-of-moments estimator whose consistency properties follow from the theory of empirical processes.

Precision
In binary classification, the fraction of positive predictions that are true positives: $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$.

Run Equilibrium
The Nash equilibrium (Definition 2.1) in which all depositors withdraw simultaneously, causing insolvency even for an otherwise solvent institution.

Social Coordination Pressure (SCP)
The BFI subcomponent capturing social-media mention velocity, sentiment polarity, and geographic concentration of concern about the institution.

Structural Mismatch Score (SMS)
The BFI subcomponent measuring the interaction between unrealised asset losses (AOCI/T1K) and liability instability.

Tier-1 Capital (T1K)
The core measure of a bank's financial strength from a regulator's perspective, comprising common equity and disclosed reserves.

The End